
Subject: Invitation to Speak at the University of Pennsylvania
Date: Monday, July 20, 2020 at 9:35:49 AM Eastern Time
From: Sarah Rottenberg
To: Somesh Rahul, Joan Kurylak
CC: Sanjeev Kumar
Attachments: img-490cdbb0-a7c5-4693-8f3d-115269236b74

Hi Somesh and Joan,

I hope you both are doing well.

Following up on our recent conversations during my consulting work with Lutron, I'd like to formally invite you both to the University of Pennsylvania to speak with our Integrated Product Design (IPD) students.

Your session on "Design Thinking in Action" would be incredibly valuable for them. We were hoping you would be available on Wednesday, October 14, 2020. The plan would be for a one-hour lecture on your work and process, followed by a one-hour portfolio review session with the students.

Please let me know if that date works for you. If not, we can certainly find another time that is convenient.

Looking forward to hopefully hosting you at Penn!

Best,

Sarah Rottenberg
Executive Director, Integrated Product Design Program
University of Pennsylvania
srottenberg@upenn.edu

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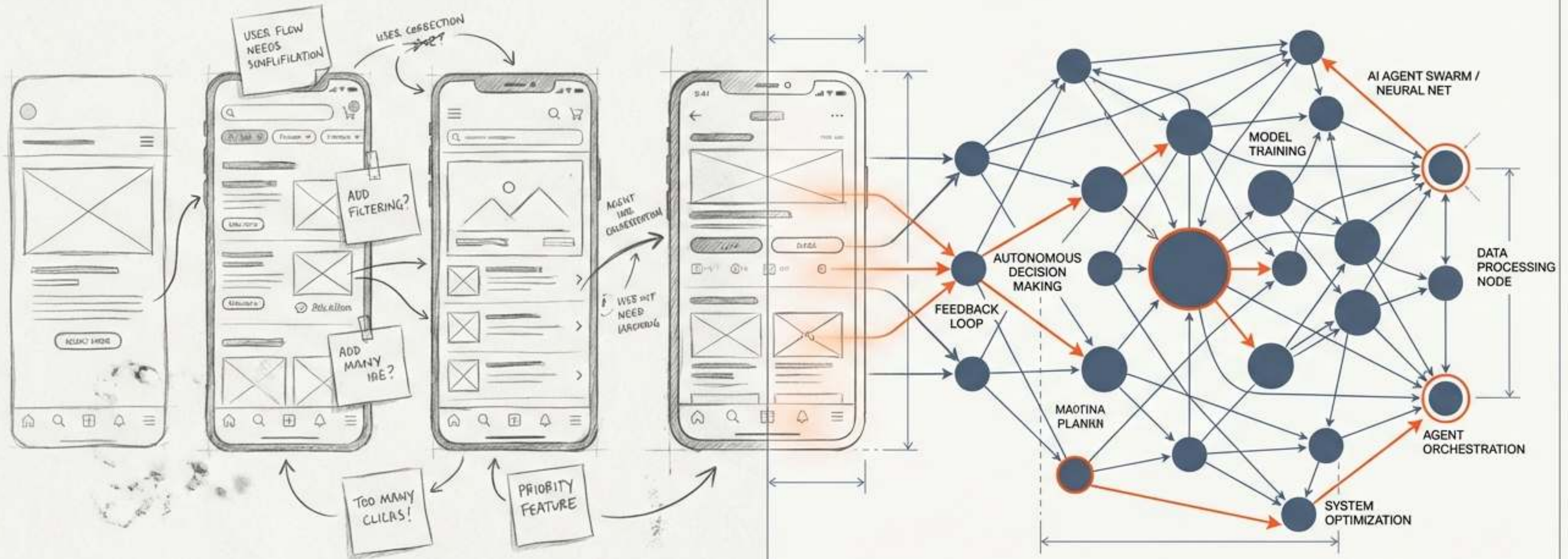
Thumbnail

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Links

Evolution of Product & Design Roles in the Age of AI

Martina Plantijn · Re-architecting the craft: From feature shipping to agent orchestration.



Martina Plantijn

GUEST LECTURE: UPENN IPD MASTERS

SPEAKER: LEAD AI PRODUCT MANAGER, EMERGING TECH TEAM, WALMART

The Great Rebuild: The 'Agent Boss' Era

Corporate Lifespan Comparison

1958 Corporate Lifespan

61 Years

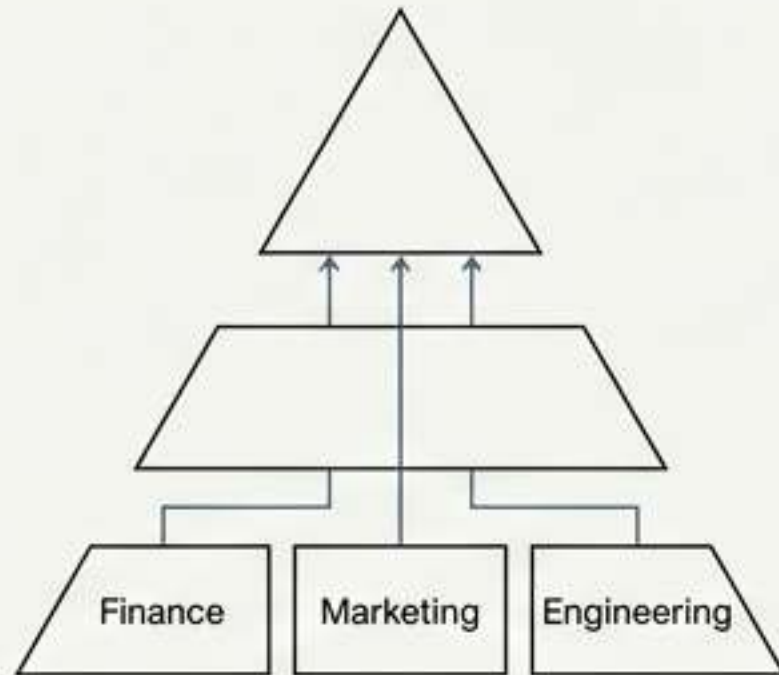
Today

<18 Years

The era of incremental change is over. We are dismantling and rebuilding the tech organization structure.

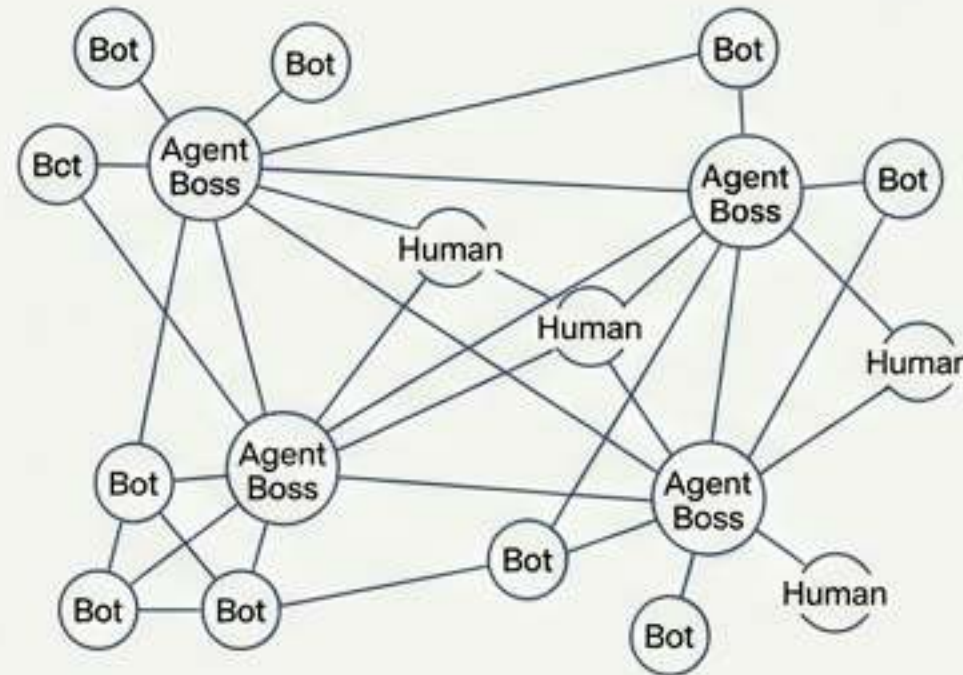
Old OS

Functional Silos



New OS

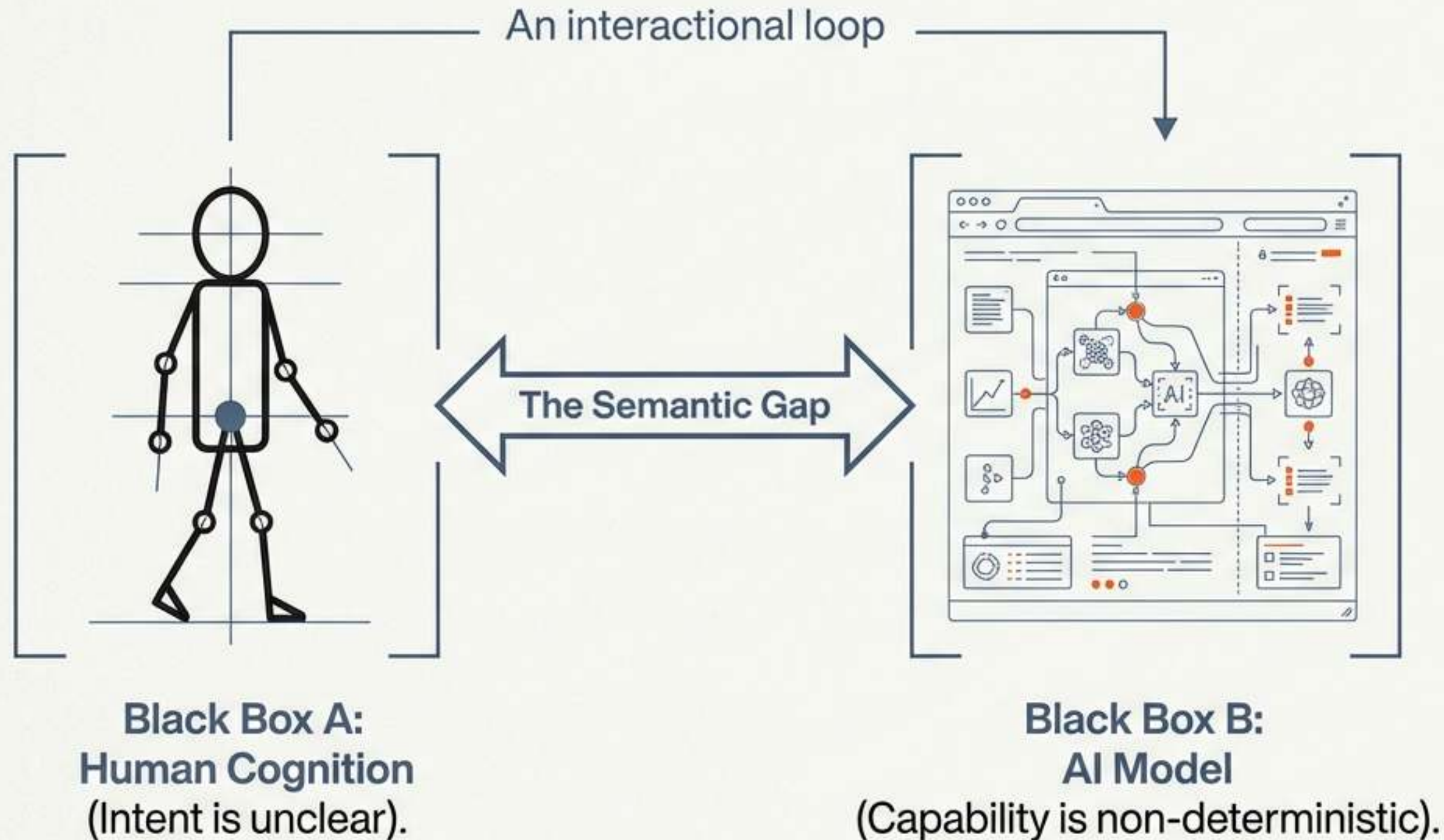
Goal-Oriented Agent Teams



Microsoft and Deloitte define the 'Agent Boss' era: A future where every employee acts as the **CEO of an agent-based startup**.

- 28% of managers plan to hire leads for hybrid human-AI teams.
- The 'Human-Agent Ratio' is the new optimization metric.
- The traditional triad (PM, Design, Eng) is evolving into a fluid ecosystem.

The Semantic Gap: Managing Two Black Boxes



From HCI to HAI (Human-AI Interaction):

The challenge is no longer just usability. It is managing the friction between two unpredictable systems.

Core Challenges (arXiv):

1. **Capability Uncertainty:** We do not know exactly what the model can do until we query it.
2. **Output Complexity:** Working with multidimensional, rich outputs rather than simple discrete states.

Death of the Static PRD

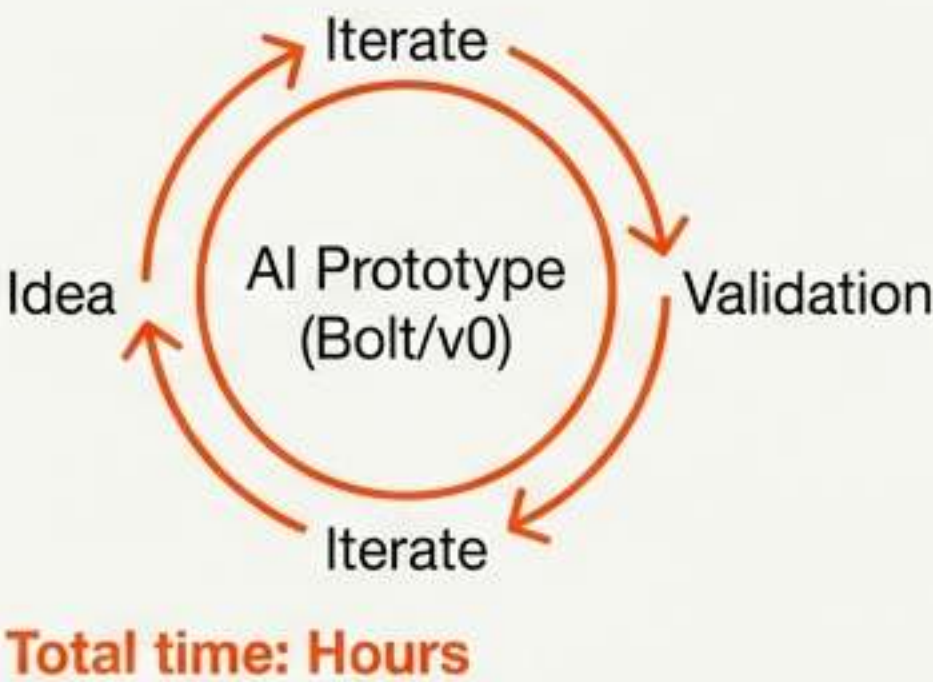
The Waterfall Spec



Workflow Shift

Writing is thinking, but **static documentation is too slow**. Engineers are persuaded by vision, not text. **The spec is the prototype**.

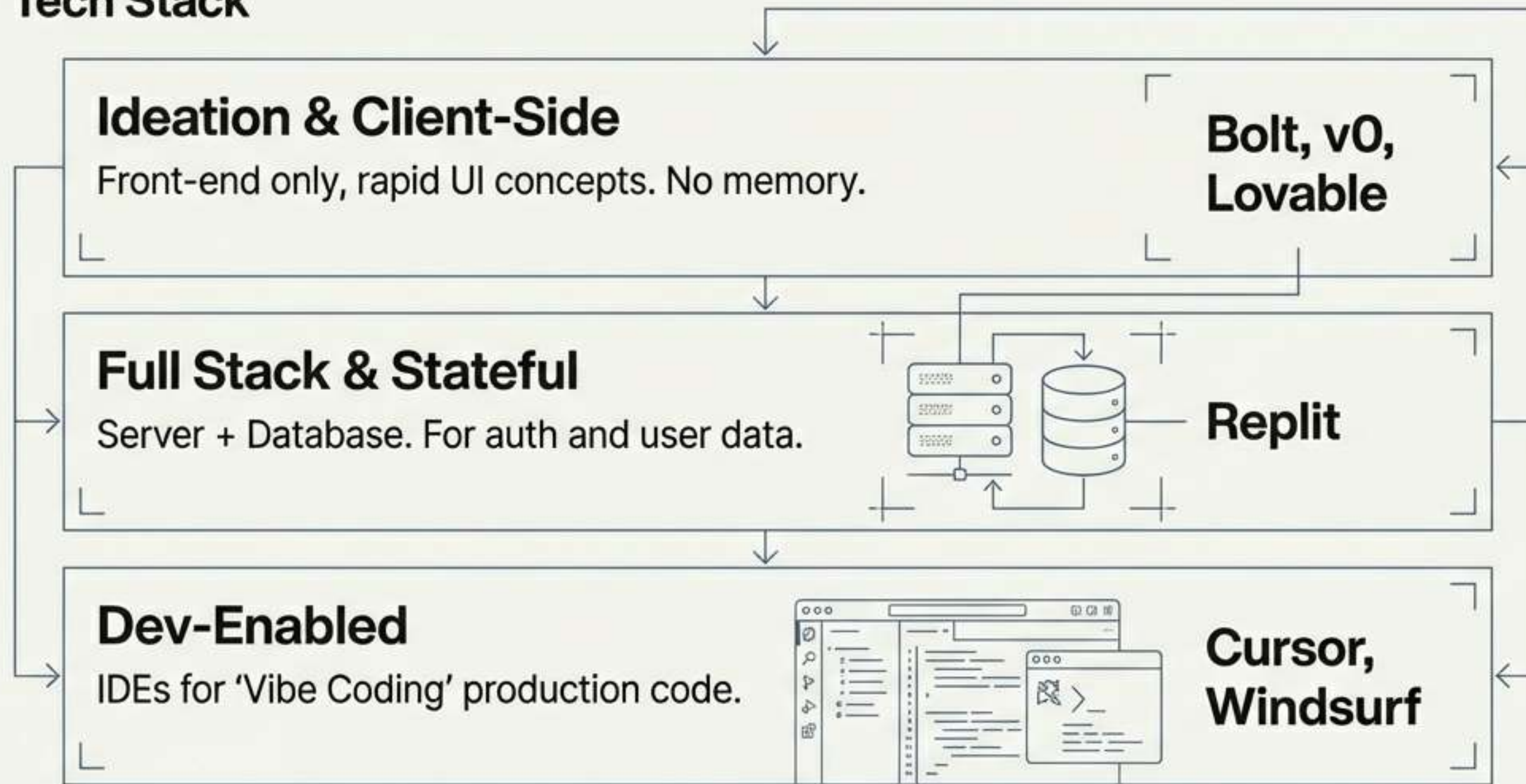
The Vibe Code



“Don’t be embarrassed to admit that you use AI... The worst generated PRD is still going to be an amazing starting point.” — Merily Nika

Prototyping is Communication, Not Code

Tech Stack



The Strategic Pivot:

You don't need to be a developer. You need to understand the stack to orchestrate to orchestrate the vision.

Goal: Have 100 more conversations with customers by showing them something real, instead of a static Figma mock.

A New Vocabulary: HAI Interaction Primitives

primitives	P: provide (X:type+, Y:type*)
types	input: raw_data fvector state query mparams ...
	output: class regression action clusters items params ...
	feedback: validate evaluate XAI.features XAI.rules ...

Application

Design Utility: Interactions are exchanges of information, not just clicks.

Example Action: **req-sample_class(X,Y)** = The model requests the user to classify a specific input. Use these primitives to map complex behaviors like Collaborative Learning or Model Contestation.

Designing for Trust: The Spectrum of Agency

Do It For Me		Do It With Me
Martina Plantijn: User delegates, Agent executes. Example: Deep Research Agents, Trip Planning.	Governance: Model Contestation We are moving from “Command and Control” to “Intent and Audit.” Designers must create UI patterns that allow humans to intervene when the AI hallucinates. Key Concept: A “Human-in-the-loop” step often exists solely because we don’t trust the model yet—but it is a necessary design constraint.	Martina Plantijn: Iterative hand-off, User & Agent co-create. Example: Vibe Coding, Collaborative Sketching.

The Rise of the 'Full-Stack' Orchestrator

92% of product leaders will own revenue outcomes by 2026.

The Generalist (Old PM)

- Managed backlog
- Wrote tickets
- Coordinated meetings
- Focus: Breadth

The Specialist (New AI PM)

- Strategic thinker
- Data literate
- Systems-level orchestrator
- Focus: Depth

Depth over Breadth: AI handles the breadth (busywork, summaries). Humans must handle the depth (**strategy, empathy, P&L**).

The 6 Traits of an AI Product Leader

Martina Plantijn, Source: Jacqueline Kunzelman, Director of AI Product at Google

01

Exceptional Product Taste

Can you spot a good idea when AI generates 100 bad ones? Taste is the only thing AI cannot simulate.

02

Visionary Leadership

Predicting the future 5 steps ahead, not just iterating on what exists.

03

Clarity in Chaos

Managing the noise of rapid model changes without creating 'thrash' for the team.

04

Compelling Storytelling

Data isn't enough; you need a narrative to sell the vision in a noisy market.

05

Full Spectrum Execution

Blurring lines between PM, Design, and Engineering.

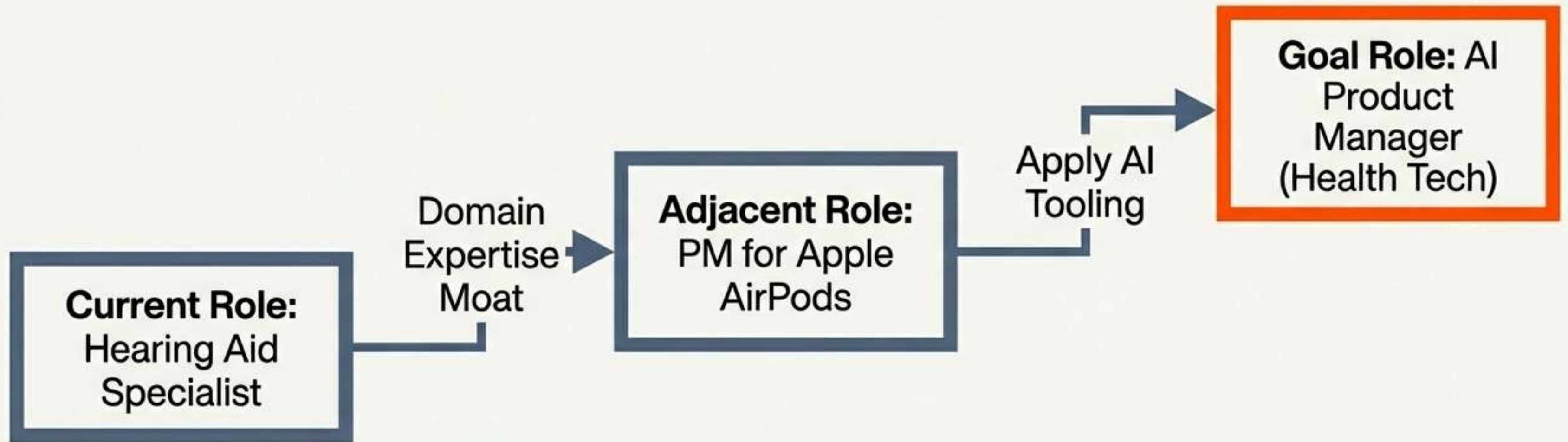
06

Deep AI Intuition

Understanding the 'why' and 'how' of models (probabilistic nature) without necessarily coding them.

The 'Crab' Career Strategy

Martina Plantijn, Source: Merily Nika



Don't apply for generic 'AI PM' roles. Apply where your *domain* meets AI. Strategy: Move adjacent (*sideways like a crab*). Your domain expertise (supply chain, healthcare, finance) is the *moat*. AI is just the tool you apply to that domain.

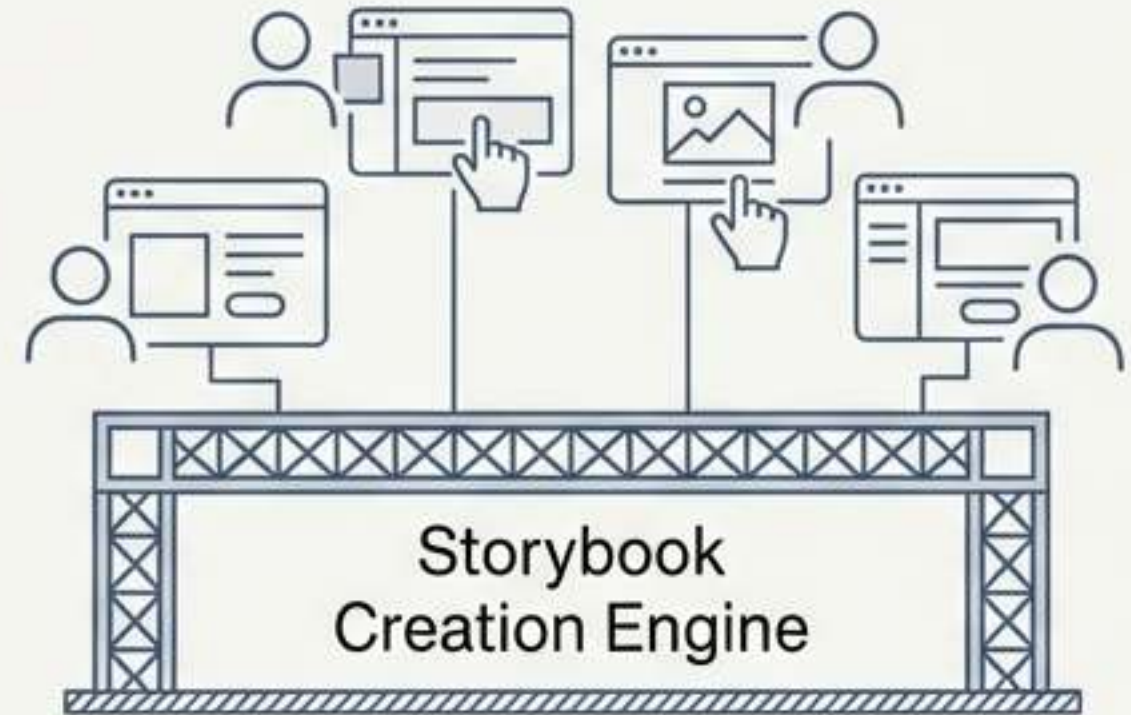
Future-Proofing: First vs. Second Order Thinking

First Order Thinking (The Wrapper)



Risk: Vulnerable to commoditization by next model update

Second Order Thinking (The Platform)

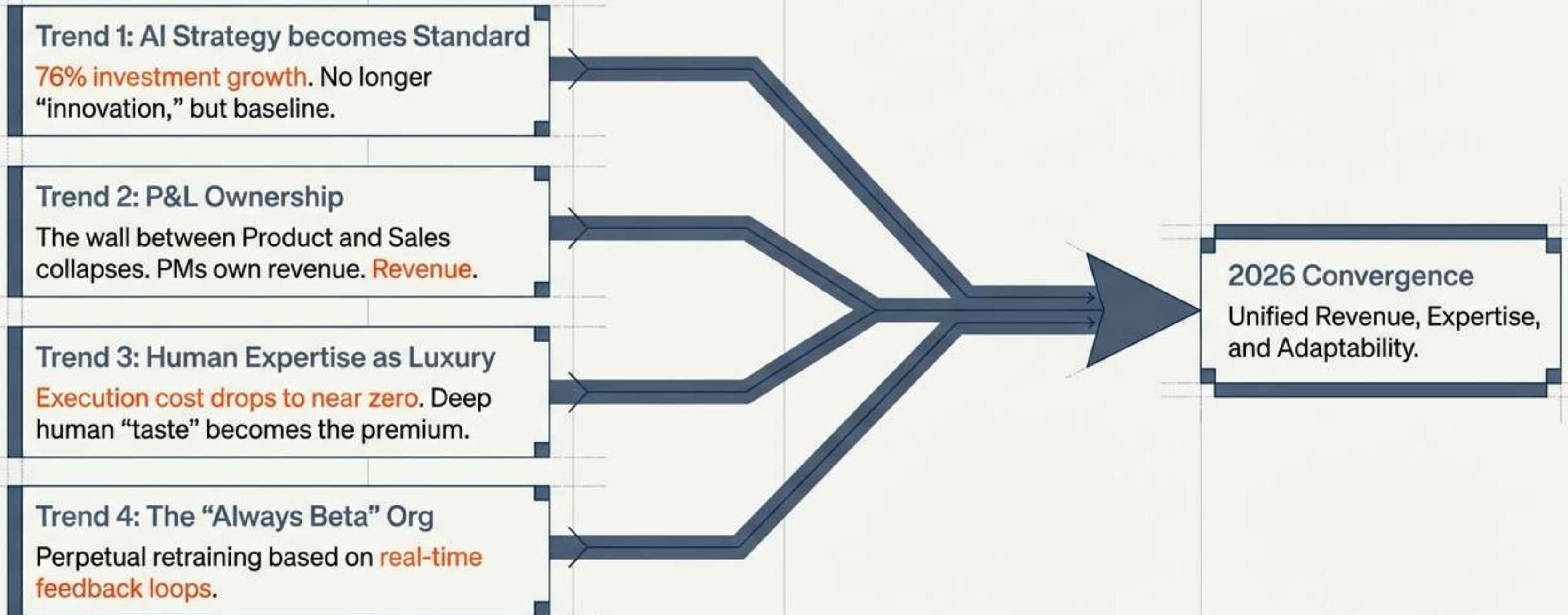


Strategy: Build platforms that enable creation.

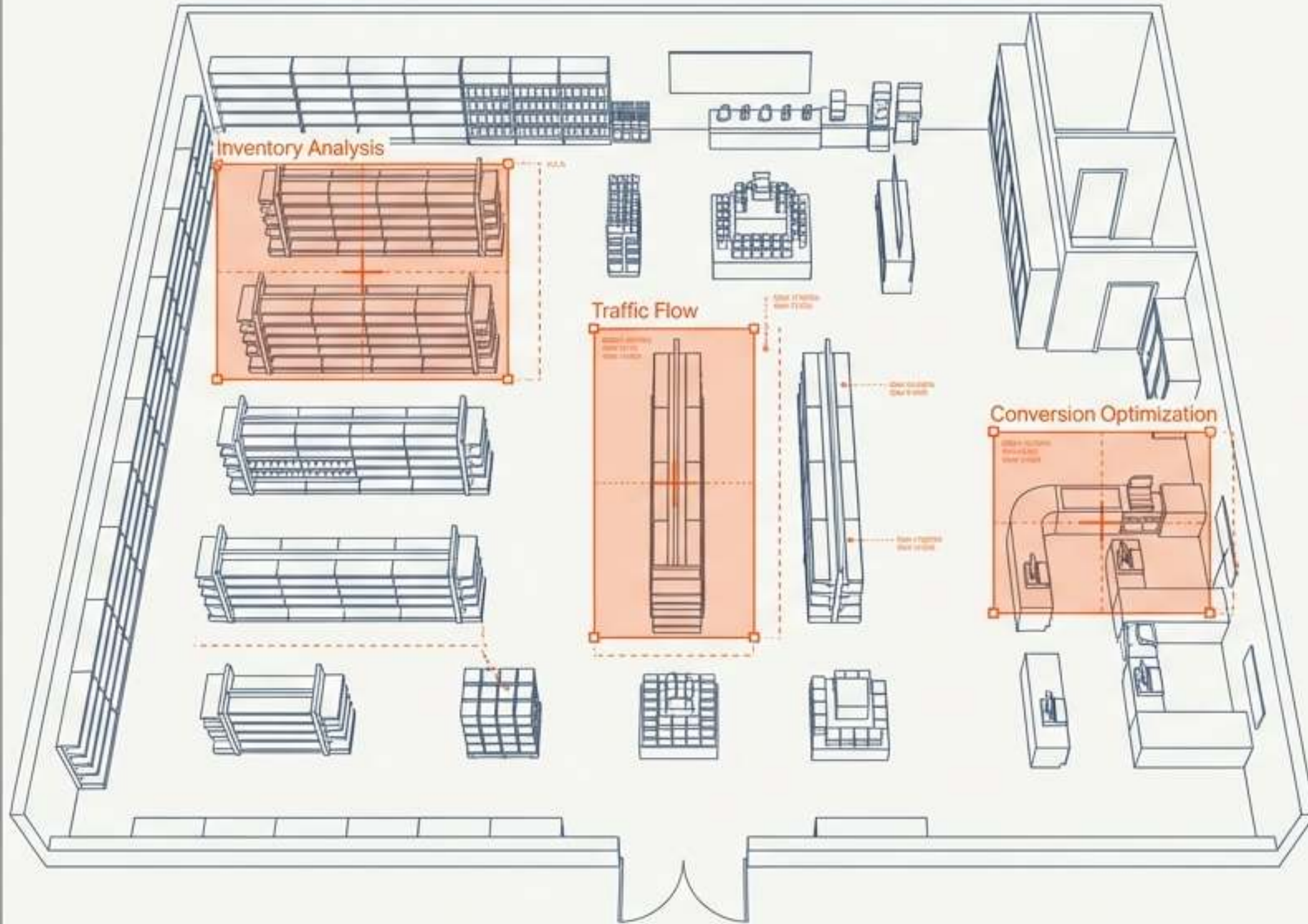
The Magic Wand Question

Ask yourself: "What human-in-the-loop step exists only because of a **technical limitation**? What would I build if that limitation disappeared tomorrow?" **Build for where the model will be in 6 months.**

The 2026 Outlook: From Features to Revenue



The Reality at Scale: Walmart Emerging Tech



We are looking for System Thinkers, not just Python coders.

The “Greek Wedding” Analogy:

“Show me any product, and I will find how it uses AI.”

Retail Application:

It’s not just security cameras; it’s optimizing layout for **revenue**. PMs must see these connections.

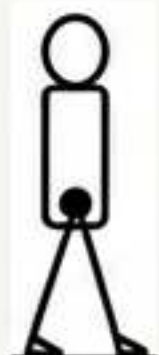
The Future is Human AND Machine

“The future isn’t human OR machine,
but human AND machine.”

Student Action Plan

- ☒ Tool Hopping: **Block 1 hour/week** for experimentation (Replit, Opal, NotebookLM).
- ☐ Don't Hide It: **Thrive** in your use of AI. **Show your work.**
- ☐ Train Your Taste: Focus on the “**why.**” **Taste** is the only thing AI cannot simulate yet.

Recommended Reading & Resources

1.  **Unpacking Human-AI Interactions**
Source: arXiv (Tsiakas & Murray-Rust) 

3.  **Real-Life Examples of AI in Design**
Source: Human Factors International

AI-Powered Design Tools
Enhance the design process with AI-powered tools that generate design concepts, optimize layouts, and streamline workflows.

Personalized User Experiences
Leverage AI algorithms to analyze user behavior and preferences, delivering tailored content and interfaces.

Predictive Analytics
Utilize AI to forecast user needs and trends, enabling proactive design decisions and feature development.

Automated Content Creation
Streamline content production with AI-generated text, images, and multimedia elements, reducing manual effort.

2.  **The Great Rebuild: Architecting an AI-native tech organization**
Source: Deloitte

4.  **Product Design & AI Courses**
Source: Aakash Gupta (YouTube / Maven)

The definitions of these roles are being written right now. **You are the ones writing them.**
— Martina Plantijn, Deep Charcoal, with **International Orange (#FF4F00)**